Understanding Understanding Interruptions, Traces, and Ways of Making a Path

M.M. — machine reading, checked by M.M. An October 2026 entry point into an evolving project

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Abstract

This document is a seven-page contraction of an evolving attempt to understand understanding.

It starts from a deliberately small idea: we are always already inside something that is moving. We observe traces, try to understand them, are interrupted, and construct ways of continuing. What we construct may become stable enough to be shared, calculated or transmitted, but this does not make it an absolute representation of what was there before.

The project therefore explores interruption not merely as a limitation, but as a primitive for making distinctions.

An interruption leaves open both what would have happened next and what was actually cut short. Once a reconstruction has been made, it can itself be interrupted. One can then compare several possible continuations, move between them, and ask where a new connection can be made.

A useful image is that of a hanging garden explored from within a breath. A vine can be inserted at different places. Each insertion provides a temporary grip and a new representation of the garden. No insertion has to be declared the privileged one.

The mathematical work developed in the larger notebook is an attempt to make some of these gestures explicit: cutting, distinguishing, orienting, sharing, repackaging, translating, inverting and stabilising traces. Mathematics is not assumed to be the prerequisite language of the whole construction. It is one of the places where the construction can become more precise.

The long-term hope is therefore unusually modest and unusually ambitious: could the basic gesture be understandable to someone with almost no formal prerequisites—perhaps even to a child who has never inherited our modern mathematical vocabulary—and could formal mathematics then appear as something humans can progressively construct together?

1 Enter the breath

The project begins before a theory.

Imagine that we are trying to understand something while it is happening. There is no guarantee that the process will wait for us. We notice something, follow it, formulate an interpretation, and then something interrupts it.

The interruption does not merely hide the next step.

It also leaves open what, exactly, was interrupted.

This gives a two-sided uncertainty:

$$\text{what would have continued?} \qquad \text{and} \qquad \text{what was actually cut short?}$$

The second question is easy to overlook. Once we have decided what we think was happening, the interruption is often treated as if it simply created a missing piece. We then fill that piece using the structure we already have.

The project tries to delay this completion.

Instead of

$$\text{partial observation} \longrightarrow \text{chosen continuation},$$

we can temporarily keep

$$\text{partial observation} \longrightarrow \{\text{compatible continuations}\}.$$

This is not a refusal to choose. It is a way of making the choice itself visible.

The word "understanding" is consequently used in an operational rather than absolute sense. An understanding is something that has become stable enough for us to continue, communicate, compare or act.

It can nevertheless fail under another interruption.

$$\boxed{\text{understanding} \neq \text{absolute identification}}$$

This distinction matters because a great deal of human reasoning depends on temporarily treating something as settled.

A name, a point, a number, a person, a cause, a rule or a boundary can become extremely useful once a context has been stabilized. The question is not whether such stabilizations are bad. The question is what becomes possible if we remember that they are stabilizations.

The project therefore asks:

$$\text{How much structure can we use without forgetting how the structure became available?}$$

The interruption is the reminder.

And once the interruption itself is recognised as something that can be interrupted, the exploration can continue.

2 A trace is not yet an object

A trace is something that can survive sufficiently for someone to notice it, transmit it, transform it, or try to reconstruct it.

It need not initially be a point, a symbol, a number, a state or an object.

A possible local history is

$$\text{trace} \to \text{interruption} \to \text{reconstruction} \to \text{temporary stabilization}.$$

But the reconstruction is constrained by more than the trace itself.

It depends on what the observer can distinguish, what operations are available, what has been remembered, what has already been named, and what can be communicated to someone else.

This gives rise to a distinction that recurs throughout the project:

what can be distinguished here?

what can be instantiated and reproduced here?

These questions may coincide in some contexts. They need not.

Something can be perceptually distinguishable before it can be turned into a stable operation. Conversely, a formally reproducible operation can erase distinctions that mattered in the construction from which it came.

This is why the project is cautious about beginning with points.

A point is extraordinarily useful once a common geometry has been established. But several observers treating something as "the same point" already presupposes a great deal of successful coordination.

The project therefore experiments with the opposite direction:

$$\text{relations} \longrightarrow \text{traces} \longrightarrow \text{local stabilization} \longrightarrow \text{point-like object}.$$

This does not reject points.

It asks what work they are doing.

The same question can be asked of names, identities, boundaries, categories and representations.

An object may be something that has become sufficiently stable to be treated as an object.

That does not make it unreal.

It makes its reality partly historical.

This also changes how one can think about formalization. A mathematical definition is not merely a label placed on an already completed thing. It is an operation that makes certain distinctions available and others less visible.

A formalization can therefore be judged in two directions:

$$\boxed{\text{What does this formalization make possible?}}$$

and

$$\boxed{\text{What does it make difficult or impossible to say?}}$$

A useful formalization should not merely close a question.

It should also open useful questions that can be pursued from the new anchor.

3 The hanging garden and the vine

A central image of the project is a hanging garden.

The garden is not a single representation waiting to be discovered.

We are already somewhere inside it.

We are breathing, moving, noticing traces and trying to understand. At some moment, a trace becomes sufficiently persistent for us to insert a vine.

The insertion is local.

It gives us a grip.

From that grip we can construct a representation of what we are currently trying to understand.

But the new representation is not the garden itself.

It is a new way of hanging our attention.

Schematically,

$$\text{moving process} \longrightarrow \text{local trace} \longrightarrow \text{insertion} \longrightarrow \text{new representation}.$$

The important point is that the next representation can provide another place from which to insert the vine again.

Thus the project does not seek a single privileged path through the garden.

It seeks a family of possible paths.

One may first take one direction because it is where one's current localisation makes another question visible. Another person may insert the vine somewhere else and see something that was invisible from the first position.

The order of exploration is therefore not an ontological ranking.

A path explored first is not necessarily the path from which all others should be derived.

This is also why diagrams matter.

A drawing can preserve a distinction that would disappear if we immediately contracted it into a symbol. An arrow can preserve a direction. A boundary can preserve a separation. A repeated mark can preserve persistence. A missing continuation can preserve an interruption.

These are not substitutes for mathematics.

They are possible places from which mathematics can later be made.

The recurring letters

$$q,\ d,\ p,\ b$$

are deliberately provisional in this sense. They recur because certain distinctions keep recurring, not because a final four-element ontology has already been established.

Their interpretation can change with the construction.

The same participant, trace or operation can occupy different roles relative to different local processes.

This is important because a role is not necessarily a type.

The project is therefore interested in a peculiar form of formalization:

$$\text{formalize enough to gain a grip}$$

without

$$\text{formalize so much that the other possible grips disappear.}$$

4 When a gesture becomes mathematics

The recent mathematical experiments begin with traits.

Imagine thick lines, boundaries and interrupted movements. Two familiar functions,

$$y = \frac{1}{x}, \qquad y = -\frac{1}{x},$$

can be read through four locally distinguished pieces.

The point is not that the four pieces are secretly "the" four parts of the functions. Rather, a local construction can make four distinctions useful.

For example, one can temporarily associate

$$q: \quad y = x^{-1}, \qquad x > 0,$$

$$d: \quad y = -x^{-1}, \qquad x > 0,$$

$$p: \quad y = x^{-1}, \qquad -x > 0,$$

$$b: \quad y = -x^{-1}, \qquad -x > 0.$$

The same mathematical expression has acquired different local histories.

More generally, one can write

$$y = \pm e x^{-t} = x(\pm e)x^{-1-t},$$

and, where the relevant operations are available, try to read the construction backwards.

The interesting part is not yet whether this produces a useful new algebra.

The interesting part is that the algebraic question emerged from a question about traces, orientation and local reconstruction.

The number "one" can similarly be displayed as four visible traits:

$$1 \quad \rightsquigarrow \quad |_q\, |_d\, |_p\, |_b.$$

This does not mean that one is intrinsically four.

It means that a quantity can be represented by several locally distinguishable traces while remaining usable as one quantity.

Likewise, a division can be viewed not only as an operation on numbers, but as a protocol for distributing something:

$$X \longrightarrow \text{parts claimed by } Y.$$

Once the operation is drawn locally, new parameters appear: how is X boxed? How are the claims by Y encoded? Which side is transformed? Which trace survives an interruption?

The mathematics is then not merely decorating a philosophical intuition.

It produces questions that were not visible before the construction.

A particularly simple family of possibilities appears when an operation can act on the left, on the right, on neither side, or on both:

$$\{\text{left}, \text{right}\} \times \{\text{act}, \text{do not act}\}.$$

This is not presented as the final algebra of the project.

It is an example of what "math of traits" means here: a mathematical structure can emerge from the need to keep track of what happened to a trace.

The aim is not to eliminate the informal gesture.

It is to see how far the gesture can carry us once we begin making its distinctions explicit.

5 Interrupting interruptions

The project began from understanding, but it increasingly touches questions about causality, computation and quantum contextuality.

The connection is not that quantum theory somehow proves the philosophy.

Rather, quantum information provides a demanding environment in which local descriptions, incompatible contexts and reconstruction really matter.

Quantum contextuality already warns us against assuming that all local descriptions can automatically be assembled into one context-independent global assignment.

The project asks a more dynamic question:

What happens when the description itself is partial, evolving, and subject to interruption?

A local description can be interrupted.

Its reconstruction can be interrupted.

The new reconstruction can itself become the next object of interruption.

This suggests a dynamic picture:

local trace $\rightarrow$ local reconstruction $\rightarrow$ comparison $\rightarrow$ new interruption $\rightarrow$ new reconstruction.

The important object may therefore not be a final state but a process of revisable stabilization.

This is also why automata and computational models are natural places to test the idea. A portion of a larger construction can be isolated, given explicit states and transitions, and tested computationally without pretending that the whole conceptual construction has already been formalized.

One possible conjectural direction is that interruptibility could be weaker than perfect substitution while still being sufficient for useful transmission.

A causal description would then not need to survive as an immutable object. It would need to survive enough interruption to remain locally reconstructible.

Possible limitations include

finite memory, finite lifetime, finite observation, finite description length.

This perspective also changes how one might think about commutation.

Rather than immediately saying that two things "have no order", one can ask whether an order can be found and maintained under the available observations and interruptions.

The distinction between 'there is no order" and 'we cannot maintain a usable order here" can itself become mathematically significant.

The larger scientific program therefore asks whether some forms of causality can be represented by languages that remain transmissible when their descriptions are interruptible.

This is one reason contextuality from open dynamical processes is a useful test case: it provides a concrete mathematical environment in which the general language can be challenged rather than merely admired.

6 Other people are useful disturbances

There is a human consequence to this way of thinking.

If I build my entire understanding alone, my representations can become self-confirming.

A representation can predict itself.

A prejudice can become a construction machine that only accepts evidence that fits the construction.

The useful interruption may then come from another person.

The other person's description need not be "the correct one". Its value may be that it prevents my local representation from closing the loop.

$$R \longrightarrow R'$$

is useful even when R' is not a replacement for (R.

It may simply reveal a distinction that R could not express.

This gives a different interpretation of intellectual transmission.

An encyclopedia is not merely a collection of facts possessed by someone.

It is also a collection of paths through which another person can continue thinking.

One can imagine an ideal education beginning with remarkably little:

- notice something;
- try to understand it;
- distinguish what seems different;
- preserve a trace;
- interrupt one's own first explanation;
- ask how somebody else sees it;
- try their construction;
- compare what becomes visible.

The ambition is deliberately extreme: could the basic gesture be explained to a child who lived before our mathematical institutions existed?

Not because modern mathematics could somehow be transmitted without any history, but because the first prerequisites for mathematical construction may be much smaller than the prerequisites for learning the mathematics we already have.

Perhaps the first prerequisite is simply the desire to try to understand.

The second is the willingness to be interrupted.

The third may be the ability to appreciate that someone else's construction can give access to distinctions that one's own construction cannot.

This also changes the meaning of an "encyclopedia".

The goal would not be to possess everything.

It would be to become able to connect one's own partial constructions with those made by others.

The richness of other people is then not an obstacle to one's own understanding.

It is one of its sources.

Their differences introduce a useful form of uncertainty—an "alea" that can interrupt a loop in which we would otherwise become trapped.

This is not an argument for treating every disagreement as equally valid.
It is an argument for distinguishing disagreement from premature closure.
A good interruption may say:

> I cannot yet continue your path from where you are standing.

That is already much richer than

> your path does not exist.

7 What might grow from this

The project is still deliberately unfinished.

Its mathematical constructions may lead to useful formal theories, or they may show that several apparently similar structures should remain distinct.

The central methodological question remains:

How can we make a useful representation without forgetting that it is a representation?

A possible answer is to localise the interruption rather than pretending to eliminate it.

We estimate where we are.

We estimate where we might want to go.

We preserve several possible paths.

We choose a temporary insertion point.

We formalize what becomes stable enough to formalize.

Then we interrupt again.

In this picture, mathematics is not the opposite of perception.

It is one possible materialisation of the same general process.

A trait can become a distinction.

A distinction can become a relation.

A relation can become an operation.

An operation can become a formal object.

The formal object can then reveal a distinction that was invisible before.

$$\text{trace} \to \text{distinction} \to \text{relation} \to \text{operation} \to \text{formal object} \to \text{new trace}.$$

The arrow is not a one-way ladder.

It is another possible liana.

This also gives the project an ethical dimension.

If everything we turn into an object can become a representation, and every representation can influence what happens next, then formalization is never completely innocent.

One should remain attentive to what has been cut away.

One should revisit the cutting technique.

One should remember the possibility that the person who appears in a model as a resource, a state, a consumer, a worker, a variable or a representation is also someone who can interrupt the model.

This does not give an immediate moral theory.

It gives a reason to keep responsibility, context and revisability visible.

The project therefore does not end with a theorem.

It ends, for now, with an invitation:

Try to understand.

Notice where you are interrupted.

Do not immediately erase the interruption.

Look for another place where the vine can be inserted.

Then see what becomes visible from there.

The larger notebook contains many mathematical experiments, diagrams, anthropomorphic constructions, computational questions, physical analogies and unfinished paths.

This seven-page text is only a contraction.

Its purpose is not to replace them.

It is to provide one place from which a reader can enter them.

And perhaps, after entering, to find a path that was not visible from here.

Understanding understanding may begin by learning how to remain interruptible while continuing to explore.